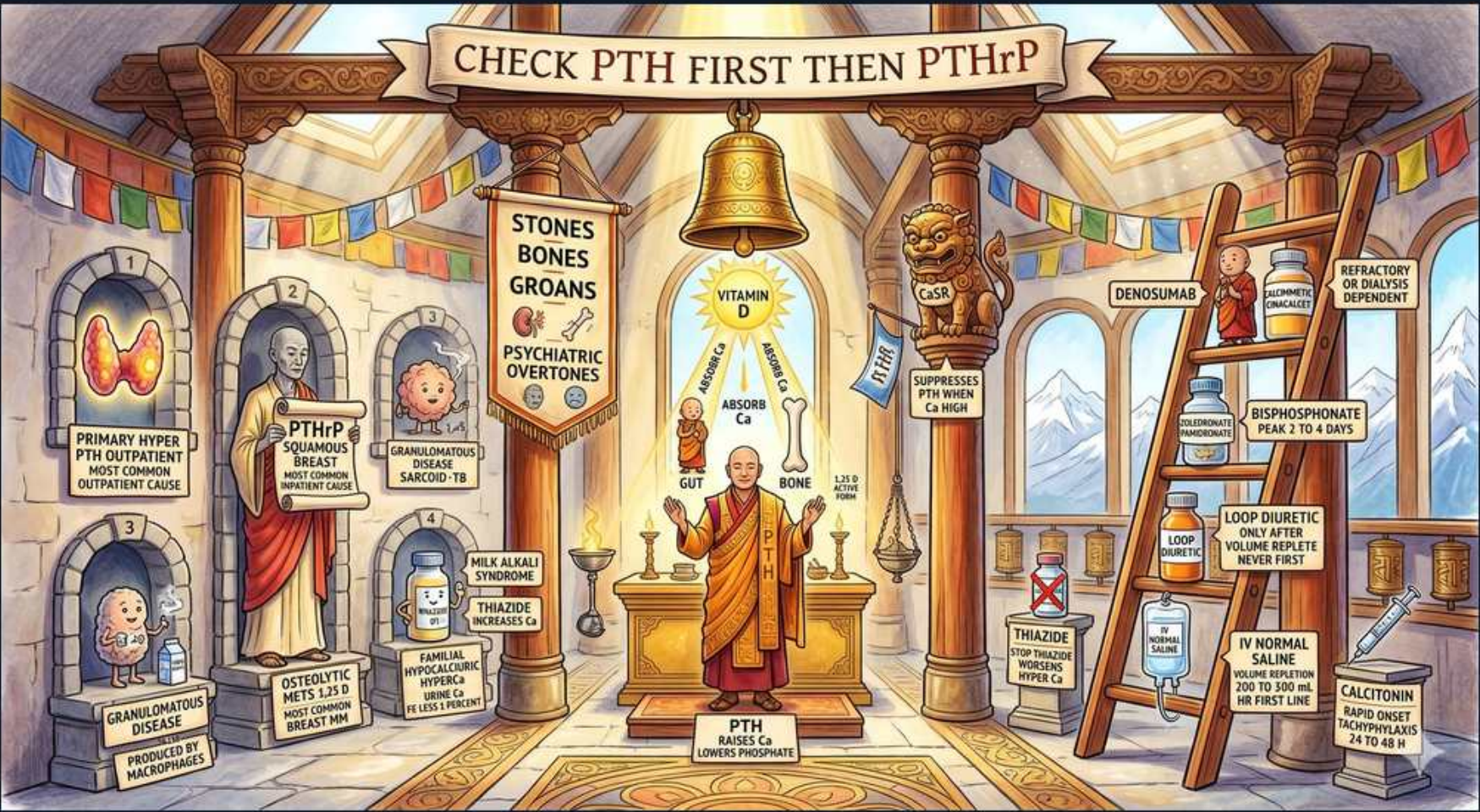


Calcium (Renal) — Hypercalcemia



1. Check PTH first

WHAT YOU SEE

Banner: CHECK PTH FIRST THEN PTHrP

WHAT IT ENCODES

The opening move in every hypercalcemia case: confirm it is real (correct for albumin / ionized Ca), then measure PTH FIRST. High/normal PTH = PTH-mediated (primary hyperPTH, FHH); suppressed PTH = PTH-independent (malignancy via PTHrP, granulomatous 1,25-D, vitamin D, etc.). Primary hyperPTH and malignancy together cause ~90% of cases.

BOARD TRAP

WRONG: ordering imaging (sestamibi/neck US) or PTHrP before PTH. PTH is the branch-point that defines the entire algorithm; everything else is downstream and only interpretable once you know whether PTH is suppressed.

2. Primary hyperPTH

WHAT YOU SEE

Panel #1: a thyroid/parathyroid statue in an arched temple niche labelled PRIMARY HYPER PTH — the carved gland marks the autonomous adenoma that is the #1 outpatient cause.

WHAT IT ENCODES

Primary hyperparathyroidism = #1 OUTPATIENT cause: an autonomous adenoma (~85%) drives high Ca, LOW phosphate, high/normal PTH, and high urine calcium. Most are asymptomatic incidental findings. Parathyroidectomy is curative; indications include Ca >1 above ULN, age <50, T-score ≤ -2.5/fracture, or eGFR <60/stones. Screen for MEN1/MEN2A if young or multigland.

BOARD TRAP

WRONG: assuming it is also the top INPATIENT cause. In a hospitalized/ill patient, hypercalcemia is malignancy until proven otherwise — primary hyperPTH dominates the outpatient/incidental setting.

3. PTHrP squamous/breast

WHAT YOU SEE

Panel #2: a robed figure with a tumor labelled PTHrP / SQUAMOUS / BREAST — the cancer figure broadcasting PTHrP is humoral hypercalcemia of malignancy.

WHAT IT ENCODES

Humoral hypercalcemia of malignancy = #1 INPATIENT cause: tumor PTHrP binds the PTH receptor → high Ca, LOW phosphate, but SUPPRESSED native PTH. Classic tumors: squamous cell (lung, head/neck), renal, bladder, and breast. It indicates advanced cancer and poor prognosis; treat the calcium emergently and the tumor definitively.

BOARD TRAP

WRONG: expecting PTH to be elevated 'because PTHrP acts like PTH.' Intact PTH is appropriately SUPPRESSED — high Ca + low PTH + low phosphate + a known cancer = PTHrP-mediated, confirmed by an elevated PTHrP level.

4. Granulomatous 1,25-D

WHAT YOU SEE

Panel: a granuloma/macrophage figure labelled GRANULOMATOUS / SARCOID — the macrophage stands for the rogue 1-alpha-hydroxylase pumping out active 1,25-D.

WHAT IT ENCODES

Granulomatous disease (sarcoidosis, TB, fungal) and some lymphomas express unregulated macrophage 1-alpha-hydroxylase → excess 1,25-OH vitamin D independent of PTH → high Ca, suppressed PTH, often hypercalciuria. It responds to GLUCOCORTICOIDs, which shut off macrophage 1-alpha-hydroxylase. Patients should avoid sunlight and vitamin D.

BOARD TRAP

WRONG: reaching for bisphosphonates first or supplementing vitamin D. Steroids are the targeted therapy for granulomatous hypercalcemia, and vitamin D/sun exposure worsens it by feeding 1,25-D production.

5. Vitamin D → gut/bone

WHAT YOU SEE

Monk with VITAMIN D, ABSORB Ca, GUT/BONE

WHAT IT ENCODES

Active vitamin D (1,25-D) raises calcium by increasing GUT absorption of Ca and phosphate and promoting bone resorption. Excess from intoxication (high 25-D) or granulomatous/lymphoma 1,25-D production causes hypercalcemia WITH hyperphosphatemia and suppressed PTH. Measure 25-D for intoxication, 1,25-D for granulomatous disease.

BOARD TRAP

WRONG: confusing the two vitamin D forms. Supplement TOXICITY elevates 25-OH-D; GRANULOMATOUS disease elevates 1,25-D. Ordering the wrong metabolite leads to a missed or misattributed diagnosis.

6. CaSR / suppresses PTH

WHAT YOU SEE

Lion statue CaSR, suppresses PTH when Ca high

WHAT IT ENCODES

The calcium-sensing receptor (CaSR) on parathyroid and renal cells senses serum calcium: when Ca is high it SUPPRESSES PTH and increases urinary calcium loss (normal negative feedback). The drug cinacalcet is a CaSR agonist (calcimimetic) that exploits this to lower PTH and Ca in hyperPTH/parathyroid carcinoma.

BOARD TRAP

WRONG: forgetting that an INACTIVATING CaSR mutation = FHH, where the receptor under-senses calcium → high Ca, high/normal PTH, and LOW urine calcium (FECa <1%) — benign, do NOT operate.

7. Milk-alkali

WHAT YOU SEE

Panel: a MILK-ALKALI SYNDROME niche showing milk/antacids — the calcium-plus-alkali jug encodes the hypercalcemia + metabolic alkalosis + AKI triad.

WHAT IT ENCODES

Milk-alkali syndrome: ingestion of large amounts of calcium plus ABSORBABLE alkali (classically calcium carbonate antacids/supplements) → the triad of HYPERCALCEMIA, METABOLIC ALKALOSIS, and ACUTE KIDNEY INJURY, with suppressed PTH. It is resurging with OTC calcium carbonate use. Treat with hydration and stopping the calcium/alkali.

BOARD TRAP

WRONG: attributing hypercalcemia + alkalosis + AKI to a primary parathyroid or malignant cause. The medication history of heavy calcium carbonate (Tums) use plus this triad points to milk-alkali — stop the supplement.

8. Thiazide

WHAT YOU SEE

Panel: a THIAZIDE diuretic niche — the thiazide is here on the hypercalcemia wall because it retains calcium (lowers urine Ca), the reason it prevents stones.

WHAT IT ENCODES

Thiazides INCREASE distal tubular calcium reabsorption → lower urine calcium and modestly higher serum calcium (which is why they prevent calcium stones). They can also unmask underlying primary hyperPTH. Hold the thiazide and recheck calcium before extensive workup.

BOARD TRAP

WRONG: swapping thiazide and loop effects. Thiazides RAISE calcium (used for stone prevention); loop diuretics LOWER calcium. 'Loops Lose calcium' — and loops are NOT first-line for hypercalcemia.

9. FHH low urine Ca

WHAT YOU SEE

Panel: FAMILIAL HYPOCALCIURIC HYPERCALCEMIA niche showing a kidney holding calcium back — the retained (low-urine) calcium is the FHH signature.

WHAT IT ENCODES

Familial hypocalciuric hypercalcemia: autosomal-dominant inactivating CaSR mutation → mild lifelong hypercalcemia with high/normal PTH but LOW urinary calcium. The discriminator is the calcium-to-creatinine clearance ratio (FECa <0.01). It is benign — no surgery, no treatment, family screening only.

BOARD TRAP

WRONG: sending an FHH patient to parathyroidectomy as if it were primary hyperPTH. Surgery fails to correct the calcium. Check 24-h urine calcium/FECa — <1% says FHH (leave it alone), >2% says primary hyperPTH.

10. Osteolytic / immobilization

WHAT YOU SEE

Panel: OSTEOLYTIC METS with breast/bone imagery — tumor eating into bone (local RANKL) releasing calcium, with PTHrP often normal.

WHAT IT ENCODES

Local osteolytic hypercalcemia: bone metastases (breast, myeloma) release calcium via local RANKL/cytokine-driven osteoclast activation — PTH and PTHrP can be normal. Prolonged IMMOBILIZATION (especially with high baseline bone turnover, e.g., Paget, young, or extensive injury) similarly liberates skeletal calcium.

BOARD TRAP

WRONG: ruling out malignancy because PTHrP is normal. Lytic mets and myeloma cause hypercalcemia LOCALLY (cytokines/RANKL), not via circulating PTHrP — pursue SPEP/UPEP and skeletal imaging.

11. Stones/bones/groans/psych

WHAT YOU SEE

List: STONES BONES GROANS PSYCHIATRIC OVERTONES

WHAT IT ENCODES

Symptom mnemonic: STONES (nephrolithiasis, nephrogenic DI/polyuria), BONES (bone pain, osteitis fibrosa cystica), GROANS (abdominal pain, constipation, pancreatitis, peptic ulcers), and PSYCHIATRIC OVERTONES (fatigue, depression, confusion, coma). ECG shows a SHORT QT interval; severe cases cause bradyarrhythmias.

BOARD TRAP

WRONG: pairing hypercalcemia with a LONG QT. Hyper-calcemia SHORTENS the QT; hypo-calcemia prolongs it. Swapping these is a classic ECG distractor.

12. IV normal saline first

WHAT YOU SEE

IV saline bag, VOLUME REplete FIRST

WHAT IT ENCODES

Step 1 of acute treatment: aggressive ISOTONIC SALINE (often 200–300 mL/h) to reverse the dehydration of hypercalcemia and promote calciuria. Hydration plus correcting volume is the foundation before any loop diuretic. Monitor for fluid overload, especially in CKD/CHF.

BOARD TRAP

WRONG: starting a loop diuretic before volume repletion. Loops in a volume-depleted patient worsen dehydration and calcium; saline comes FIRST, loops only later if the patient becomes volume overloaded.

13. Calcitonin rapid

WHAT YOU SEE

Calcitonin bottle, rapid onset 24–48h

WHAT IT ENCODES

Calcitonin (4 IU/kg SC/IM) inhibits osteoclasts and increases urinary calcium — the FASTEST agent, lowering Ca within hours. But it is short-lived: TACHYPHYLAXIS develops in 48–72 h. Use it as a BRIDGE alongside saline while the bisphosphonate (slow onset) takes effect.

BOARD TRAP

WRONG: relying on calcitonin as monotherapy/durable control. It loses effect within 2–3 days; it must be paired with a bisphosphonate or denosumab for sustained calcium lowering.

14. Bisphosphonate/denosumab

WHAT YOU SEE

Denosumab + bisphosphonate ladder, peak 2-4 days

WHAT IT ENCODES

For durable control: IV bisphosphonate (zoledronic acid 4 mg, peak effect 2–4 days) is first-line for malignant hypercalcemia; DENOSUMAB (anti-RANKL) is preferred in renal failure (no renal dosing) or bisphosphonate-refractory disease. Both lower bone resorption; watch for hypocalcemia, jaw osteonecrosis, and (bisphosphonate) renal toxicity.

BOARD TRAP

WRONG: giving IV bisphosphonate in significant renal impairment or expecting immediate effect. Onset is 2–4 days (hence the calcitonin bridge), and in renal failure denosumab is the safer antiresorptive.

15. Loop only if overloaded

WHAT YOU SEE

Loop diuretic panel, only after volume replete, never first

WHAT IT ENCODES

Loop diuretics (furosemide) promote urinary calcium loss but are reserved for patients who become VOLUME OVERLOADED after saline (e.g., CHF/CKD) — never as initial therapy. Routine loop use in a dehydrated patient is harmful. Saline + antiresorptive is the real backbone.

BOARD TRAP

WRONG: choosing 'furosemide' as the first-line treatment for hypercalcemia. This outdated answer is a trap — loops are adjuncts only after volume repletion in fluid-overloaded patients.
